

LAB EXERCISE - 9

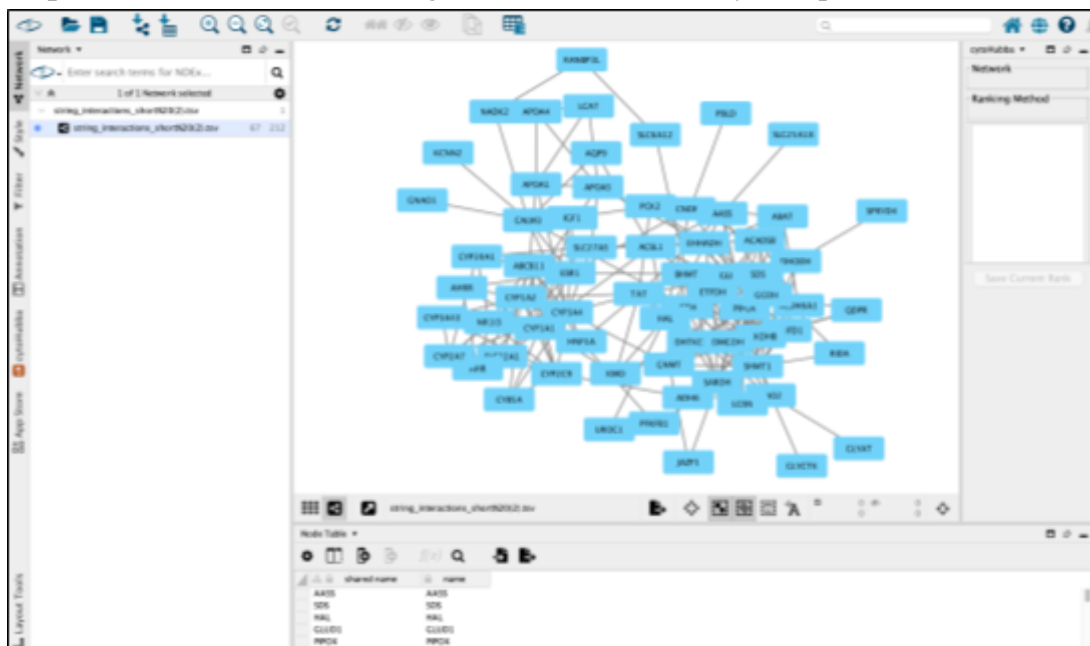
MCODE and Cytohubba

Aim: To identify highly interconnected regions (clusters/modules) within a protein–protein interaction (PPI) network using the MCODE plugin in Cytoscape.

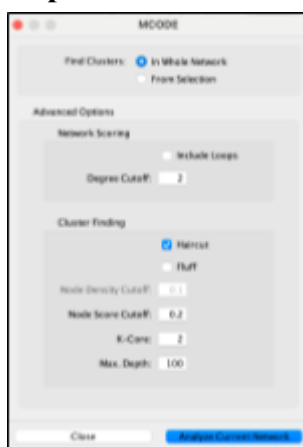
Take the network obtained from the STRING database, visualize it in Cytoscape, and use MCODE to identify clusters and also use Cytohubba plugin in Cytoscape use identify the top Hub genes.

Methodology:

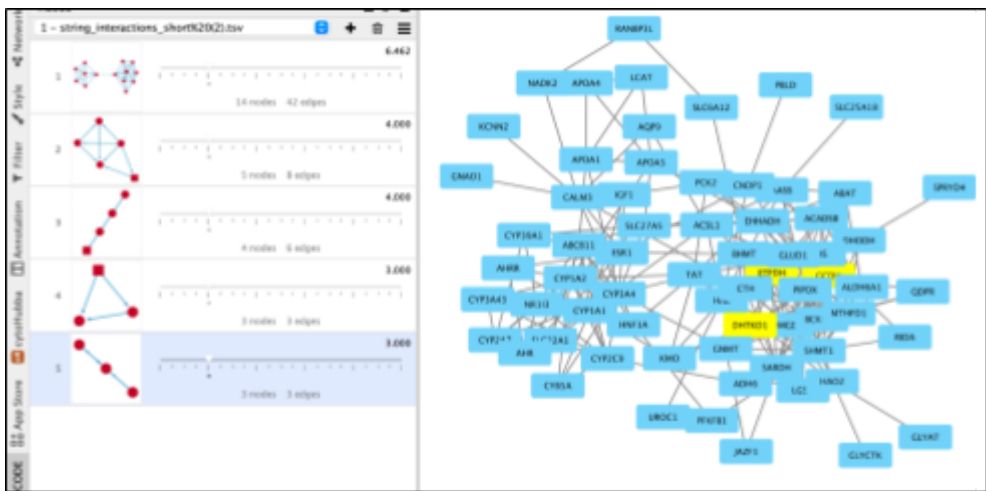
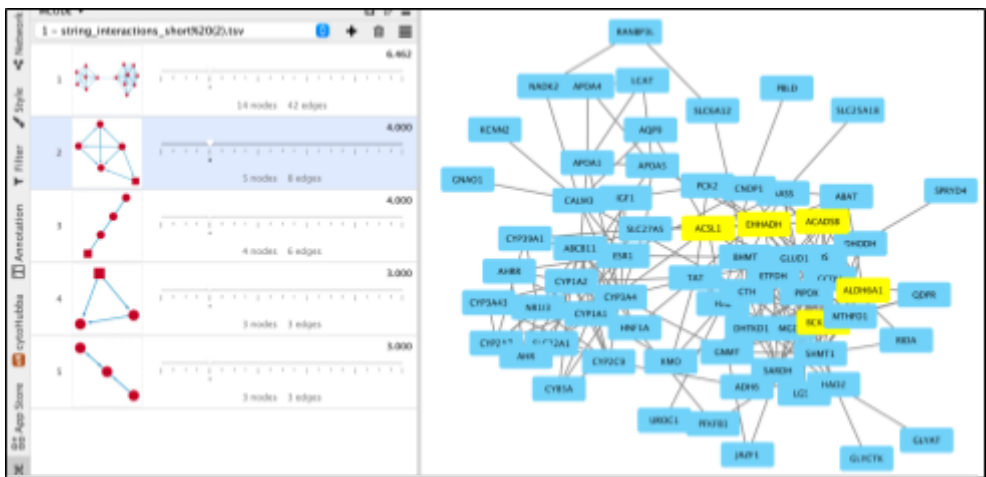
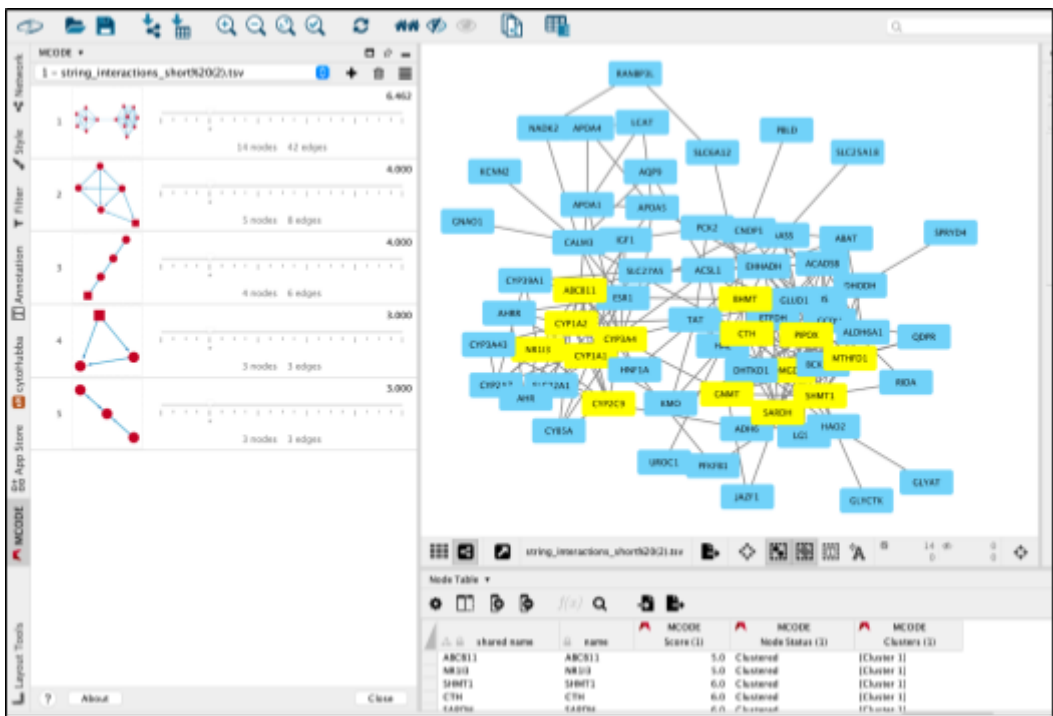
Step 1 - Visualize the Network got from STRING in Cytoscape.

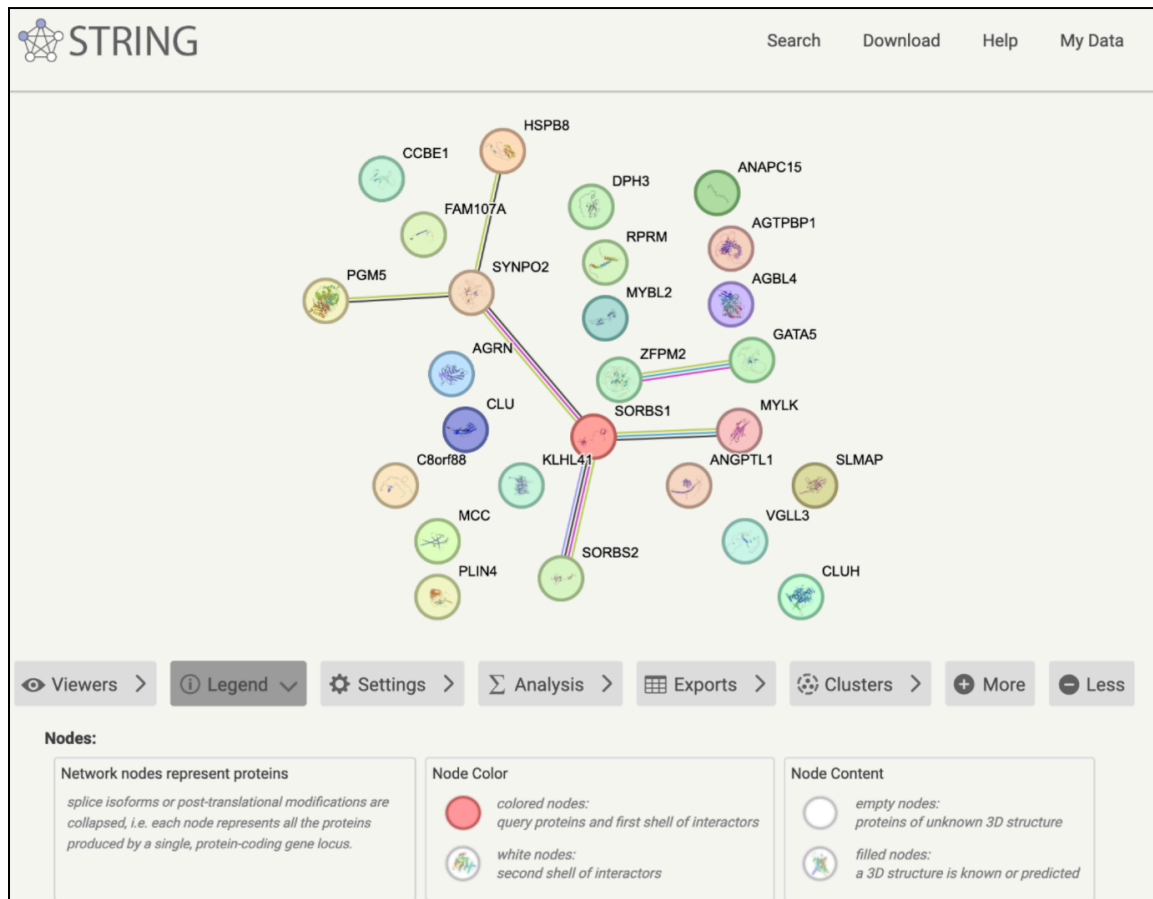


Step 2 - Click MCODE plugin, check the parameters and click Analyze Current Network.

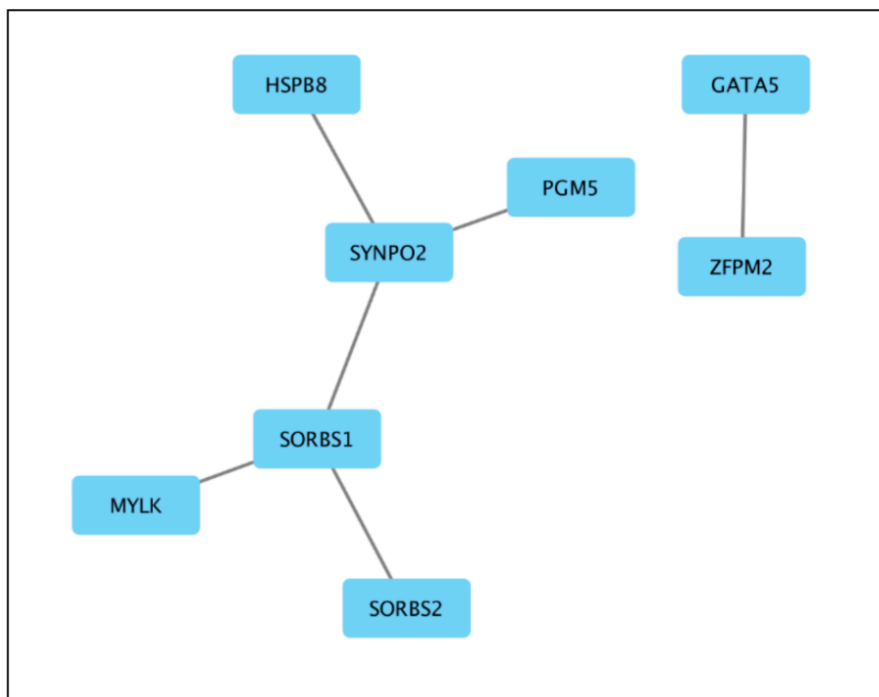


Step 3- Now, different clusters of the same set of proteins can be seen

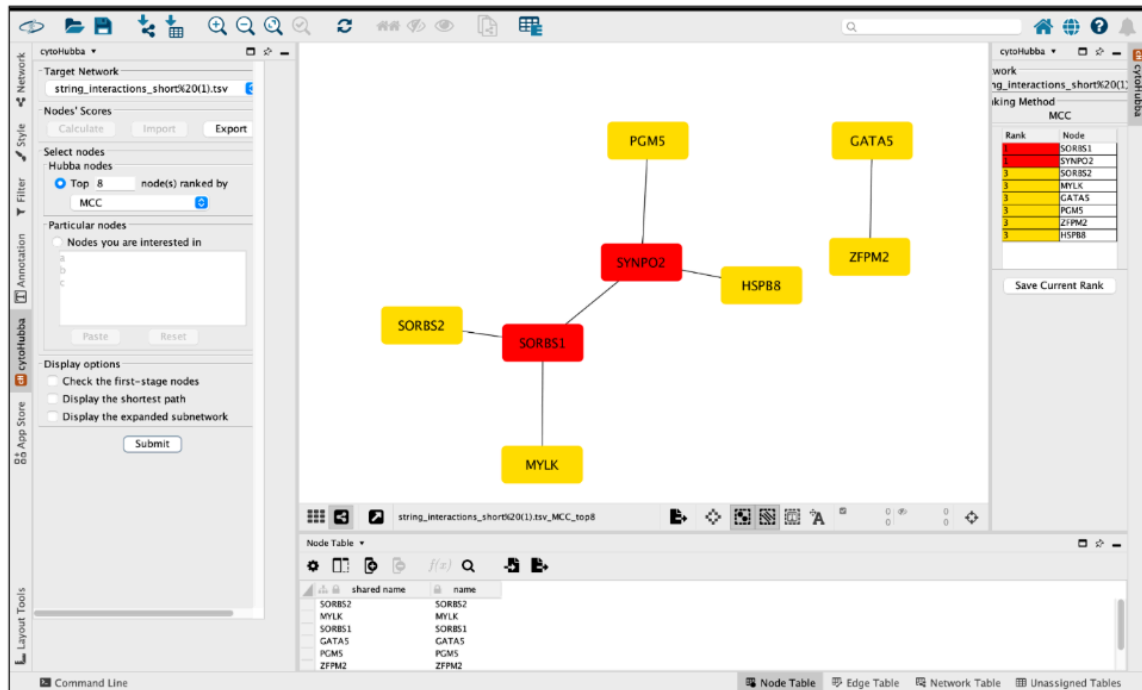




Step 4 - Download Cytoscape and open the Software then click Open Session to visualize the Network from STRING.



Step 5 - After this HUB genes identification can be done using Cytohubba plugin in Cytoscape.



Results: Cytoscape was successfully explored and one of its plugin is used to identify the Top HUB genes using Cytohubba Plugin. Here, **SYNPO2** and **SORBS1** are the top 2 Hub Genes.